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Optimizing HVAC systems for semiconductor fabrication: A data-intensive framework for energy efficiency and sustainability

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ABSTRACT

This paper presents a detailed comparative analysis of the performance and energy consumption of heating, ventilation, and air conditioning (HVAC) systems as standalone units and integrated systems, specifically focusing on semiconductor manufacturing facilities (SMFs). Highlighting the urgent need for HVAC system optimization to minimize energy waste and reduce greenhouse gas emissions, the study uncovers distinct energy consumption patterns across residential, commercial, and industrial sectors, emphasizing the unique needs of SMFs. It reveals the need for more current energy codes and standards in addressing the specialized energy demands of semiconductor fabrication plants (fabs), thereby advocating for a customized HVAC energy-design framework to enhance energy efficiency within these facilities. The research is divided into three main segments: (1) identifying the key factors that drive HVAC systems' energy consumption, (2) examining the factors that influence system performance, and (3) analyzing how these factors impact the optimization of HVAC systems in SMFs. By developing a framework that integrates design, engineering, and energy consumption data, the paper lays the groundwork for a data-intensive design approach tailor-made to meet the energy efficiency requirements of semiconductor fabs. The study's pivotal findings highlight the deficiencies of existing energy codes and standards for SMFs and propose a bespoke HVAC energy design framework. This strategy identifies critical energy consumption and performance factors unique to SMFs, recommending a data-driven design method for enhanced energy efficiency. This forward-thinking approach aims to significantly reduce energy waste and greenhouse gas emissions, establishing a new benchmark for sustainable practices in semiconductor manufacturing.

Nomenclature

ACH	Air Change per Hour
AHU	Air-handling unit
AI	Artificial Intelligence
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
BF	Building factor

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BIM	Building Information Modeling
BTO	Building Technologies Office
BTU	British Thermal Unit
CAV	Constant air volume
CFM	Cubic Feet Per Minute
CMP	Chemical-mechanical polishing
dB	Decibel
DCIMO	Design, Construction, Installation, Maintenance, and Operation
DOE	Department of Energy
ECF	Energy conversion factor
EF	External factor
EIA	Energy Information Administration
EMI	Electromagnetic Interference
ENGF	Engineering factor
EPM	Energy Performance Metric
EPA	Environmental Protection Agency
ERS	Energy Resource Station
FS	Federal standard
GPM	Gallons Per Minute
HVAC	Heating, ventilation, and air conditioning
IECC	International Energy Conservation Code
ISO	International Organization for Standardization
MW	Megawatt
RCI	Residential/commercial/industrial
RH	Relative humidity
SMF	Semiconductor manufacturing facility (fab)
TSPR	Total System Performance Ratio
VAV	Variable air volume

1. Energy consumption patterns influenced by regional weather, occupant behavior, and building functions

HVAC systems provide thermal comfort and healthy indoor environments for occupants and support manufacturing. HVAC systems have significant energy in any building type, but the energy demand for HVAC systems differs between industrial and residential/commercial buildings [1–3]. Tables 1A–1C summarize previous studies on the average energy consumed by HVAC systems.

Prior studies shown in Table 1 indicate that the proportion of energy consumed by HVAC systems over the total energy consumed by buildings fluctuates by building type and region. In the same area, for example, in Saudi Arabia (Dhahran), HVAC systems for resi-

Table 1A
Energy consumption of HVAC systems in commercial and residential buildings.

Total Energy consumed by HVAC over the whole building	Locations, building types, data types, and descriptions
50 % [1,4–7]	Developed countries (United States, etc.). Residential.
31 % [8]	United States. Residential buildings. Present the research results at the ERS of the Iowa Energy Center in Ankeny, Iowa, designed to evaluate and demonstrate commercial HVAC systems in a typical office-type facility.
22.4 % (total energy consumed in the United States in the residential sector) [9]	HVAC represents the most significant. Propose a spartan, physically based model to simulate and assess HVAC energy consumption.
11 % [10]	Europe. Residential. Propose an approach to optimize the design and operation of HVAC systems.
18 % [11]	Libya. Residential.

Table 1B
Energy consumption of HVAC systems in commercial and residential buildings.

Total Energy consumed by HVAC over the whole building	Locations, building types, data types, and descriptions
7.4 % in cold climate (reduction in energy use) to 12.7 % in hot climate (increase in energy use) [12]	Stockholm (cold), Istanbul (mild), and Doha (hot). Residential. The data shows the change in the annual driving energy of the HVAC system.
60 % [13]	Commercial (hospitals).
35 % [14]	United States. Residential and commercial.
40 % [15]	United States. Commercial. Present a typical office building on the USC campus in Los Angeles, California, to validate HVAC system energy efficiency.
46 % (Airport A), 63 % (Airport B), and 60 % (Airport C) [16]	Greece. Commercial (three airports: A, B, and C). Heating, ventilation, and cooling percentages of all energy consumption.
57.8 % (Miami), 68.7 % (San Francisco), 31.4 % (Chicago), and 19.0 % (Fairbanks) [17]	United States (Miami, FL; San Francisco, CA; Chicago, IL; Fairbanks, AK). Residential. Represent the user group with excellent environmental awareness.
48.5 % (Athens) and 32 % (Thessaloniki) [18]	Greece (Athens and Thessaloniki). Residential and commercial (office, educational, etc.). The energy required for cooling and dehumidifying the ventilation air has increased constantly through the last decades.

Table 1C
Energy consumption of HVAC systems in commercial and residential buildings.

Total Energy consumed by HVAC over the whole building	Locations, building types, data types, and descriptions
25–30 % [19]	Saudi Arabia (Dhahran). Commercial. Indicate how much energy can be saved by correctly selecting and operating HVAC systems.
73 % [20]	Saudi Arabia (Dhahran). Residential. Represent the air-conditioning of the total energy.
Almost 50 % or less [21,22]	Depending on the location. Commercial. Represent the energy to support indoor thermal comfort conditions.
60 % [23]	Saudi Arabia. Residential. Represent air conditioning systems.
16–50 % (worldwide) [24]	Depending on the country. Residential.
30 % [25]	Spain (Donostia-San Sebastian). Commercial (office).
40 % [26]	United States. Commercial. Represent the amount of energy demand at least consumed by HVAC.

dential buildings consume significantly more energy than commercial buildings. The power consumed by HVAC systems in residential buildings is far lower in European countries than in Saudi Arabia and the United States. It is well-researched that energy consumption in residential/commercial buildings is influenced by regional weather, occupant behavior, and building functions.

2. Energy consumption of semiconductor fabrication plants (fabs)

Unlike residential/commercial buildings, the energy consumption of industrial buildings is influenced heavily by the manufacturing processes and not by the occupants. The proportion of energy consumed by the occupants versus the production processes is insignificant. Especially in SMFs that require particulate-free air at the right temperature and moisture content, the power consumed by HVAC systems is far more extensive than residential/commercial buildings - (1) Significant amounts of heat are generated during the manufacturing process and high temperatures over 1000 °C can be reached for the cutting of semiconductor chips [27,28]; (2) relative humidity needs to be maintained at around 50 % throughout the manufacturing process to prevent electrostatics [29]; and (3) air contains less than 0.5 mm per cubic meter of particles [30–33]. Additional energy is required to force air through a highly high-quality air filtration system to remove particles as small as 0.3 μm [34], the constant reheating and re-cooling of air, constant removal and injection of humidity, and removal of extreme heat from the production system. The temperature, moisture, and air quality needed by chip production far surpasses the need for occupants; thus, occupants' needs become insignificant.

Building function dictates energy consumption patterns, while HVAC energy consumption patterns seem to be similar among fabs as Table 2 shows a typical range of 30–45 % of the percentage of energy consumed over the total energy consumed by HVAC systems in SMFs, with two exceptions of 67 % in California [38,39,41–43].

One research suggested that operations, purposes, and functions are the critical reasons for factors to be different for different building types. Prior studies from the EIA showed that buildings managing bulk chemical processes had similar energy consumption patterns as SMFs, suggesting that SMFs might share similar HVAC systems' energy consumption patterns with chemical plants [45].

Cleanrooms in SMFs can be designed, constructed, maintained, and operated using insights from analyzing residential and commercial cases, ensuring energy efficiency and compliance with energy codes and standards. Identifying distinct energy consumption patterns across residential, commercial, and industrial sectors, the study emphasizes the need for HVAC system optimization to minimize energy waste and greenhouse gas emissions.

3. Research objectives and methodology

Prior research suggested that the design, operation, and management of HVAC systems in SMFs differ significantly from those used in other manufacturing processes [46]. Operation and system efficiency and optimization are critical to chip manufacturing. At the same time, the need to change airflow, temperature, and humidity becomes less necessary as the demand for high-quality air remains the same [47,48]. Chip manufacturing is a 24/7 operation, and demand for high-quality air does not change. The process requires removing heat of over 1000 °C, toxic gases, and nanoparticles, maintaining air humidity, and critical functions running continuously 24/7 in SMFs.

Table 2
Energy consumption of HVAC systems in SMFs and the conditions behind consumption.

HVAC Systems Energy Proportion	Conditions
30 % (fab 1) and 36 % (fab 2) [35]	China (Zhuzhou, Hunan Province). Represent the HVAC energy consumption (refrigerant plant and circulation water systems account for the majority) based on EPMs in two fabs.
40%–45 % [36]	United States. Represent the HVAC equipment accounts of the total operating costs.
38 % [37]	Represent a typical proportion of energy consumption for HVAC systems in a fab.
36–67 % [38]	California. Represent the HVAC energy use accounted for the total cleanroom energy use in three facilities.
27.2 % [39]	Taiwan. Represent the energy consumption of chillers, part of HVAC systems that have cooling capacities of HVAC systems much more extensive than those of ordinary commercial buildings [40].
30%–75 % [40–43]	Represent the HVAC system of the total energy used to control the indoor environment in the cleanrooms. A new calculator for ECFs was developed to analyze the energy consumption in Taiwan in detail.
33 % [44]	Taiwan. Represent HVAC for cleanroom as the most energy-intensive system in a 12″ semiconductor fab based on a survey analysis.

3.1. The focus of the research

The primary research focuses on (1) understanding the factors driving the energy consumption of SMFs; (2) understanding the energy design methods, including relevant codes, standards, and best practices for SMFs; and (3) determining the data needs for SMFs' HVAC systems design. Building energy codes, standards, and best practices, such as ASHRAE 90.1, are used to improve building energy efficiency [49]. In addition to energy efficiency codes and standards, SMFs must conform to cleanroom standards such as U.S. Federal Standard 209E (FS 209E) and ISO 14644-1/2 [50,51].

Balancing between HVAC systems optimization and energy efficiency is critical to ensure efficient chip production while maintaining energy efficiency. The research focuses on understanding and identifying the gaps between design, installation, construction, operations and replacement, and interconnectivity to integrate optimal and efficient operation. The study covers, first, relevant codes, standards, and best practices; second, factors driving production optimization and energy efficiency; and third, the synergy and legacy of energy efficiency design for SMFs' HVAC systems.

3.2. Research methodology

SMFs can optimize HVAC systems by integrating data, engineering principles, and artificial intelligence. The methodology consists of the following components: First, identifying factors that affect HVAC energy consumption and performance; second, analyzing factors that influence system performance and energy consumption in SMFs; and finally, developing a data-intensive design framework for SMFs. This methodology aims to enhance energy efficiency, optimize system performance, and provide insights for future research and practical applications related to SMFs.

4. Factors affecting energy consumption of buildings

Prior research established the control factors impacting energy consumption in most buildings as listed in Table 3 [52], industrial buildings in Table 4, and SMFs in Table 5.

Energy consumption characteristics of industrial buildings are similar to SMFs but very different from residential/commercial buildings, as the production processes, and not occupants and appliances, are more important drivers of energy consumption in industrial buildings [69]. The energy demand for SMFs comes from the heat generated by the lithography, electroplating, and wafer CMP processes [70], high air quality standards, and constant 50 % atmospheric relative humidity. Tables 4 and 5 highlight the differences

Table 3
Primary factors driving building energy consumption.

Primary Factors	Control Factors
Building	Size, orientation, and design of a building [53].
HVAC systems	Design, construction, control, operation, and management [54].
Lighting	Types, control, and management [55].
Appliances and electronics	Types, usage, and management [56].
Occupancy	Types and total number [57].
Weather and climate	Weather patterns and local climate conditions [58–60].
Building types	Functions, purposes, and design.

Table 4
Several factors that drive energy consumption are related to factory production facilities.

Factors	Control Factors
Production processes	Equipment, appliances, and tools used for production [61].
Equipment efficiency	Design, types, construction, control, and management [62].
Lighting	Types, control, design, and management [55,63].
HVAC systems	Design, installation, operation, maintenance, and management [54].
Building design	The factory building's size, type, orientation, and design can impact energy consumption [64,65].
Off-peak/idling	Policies and management methods [66].
Energy management practices	Practices, technologies, monitoring, policies, and management methods [67,68].

Table 5
Several factors drive energy consumption in SMFs.

Factors	Control Factors
Energy-intensive processes	Equipment like high-temperature annealing, lithography, electroplating, and wafer polishing [28,73].
Equipment efficiency	Equipment specifications and optimization methods [74].
HVAC systems	Design, equipment efficiency, control, and management methods [75,76].
Building design	Size, orientation, and design [77].
Off-peak/idling	Policies and management methods [78].
Energy management practices	Practices, technologies, monitoring, policies, and management methods [79].
Cleanroom requirements	Air temperature, quality, humidity, and quantity [28,75].

between the control factors of most industrial buildings and SMFs, as SMFs' HVAC systems consumed significantly more energy than other industrial buildings [71,72]. The literature review affirms the critical differences between SMFs and other buildings.

- Building orientation and façades are not crucial in fabs, as literature reviews indicate. The façades are mostly thick layers of concrete and insulation. Thus, thermal transmittance through the façade is minimal. There are few windows in all the fabs except the attached office buildings or spaces. However, the footprint of the office space is less than 5 % of the fabs and does not consume much energy.
- Occupants, appliances, and equipment in SMFs represent an insignificant energy consumption compared to the chip manufacturing process, and energy consumption variability due to changes in building occupancy becomes insignificant.
- Effects of weather and climate are less significant in SMFs than in residential, commercial, and industrial buildings. The thick façade and the insignificant energy consumed by occupants, office appliances, and equipment do not create significant differences from the changing outdoor environment. While the weather would affect the need to treat outdoor air, a substantial quantity of the air is recycled as oxygen depletion is insignificant.

The energy-consumption management and control of SMFs are summarized in Fig. 1.

The energy consumed by SMFs can be divided into two main drivers: the contents (equipment and supporting services) and the functions and purposes of the contents. These components' design, operation, maintenance, and management are critical, as the production processes and auxiliary equipment supporting the procedures must be optimized before considering energy efficiency.

5. Comparison of the functions of HVAC components in SMFs and RCI

A typical HVAC system consists of a heating unit, cooling unit, humidifier, dehumidifier, filters or filtration system, air extractor, ducts, and more sophisticated control equipment (such as thermostat and automatic register) added to enhance the system control. Fig. 2 illustrates the ducting system in the fab cleanroom under installation. System control equipment includes sensors, airflow con-

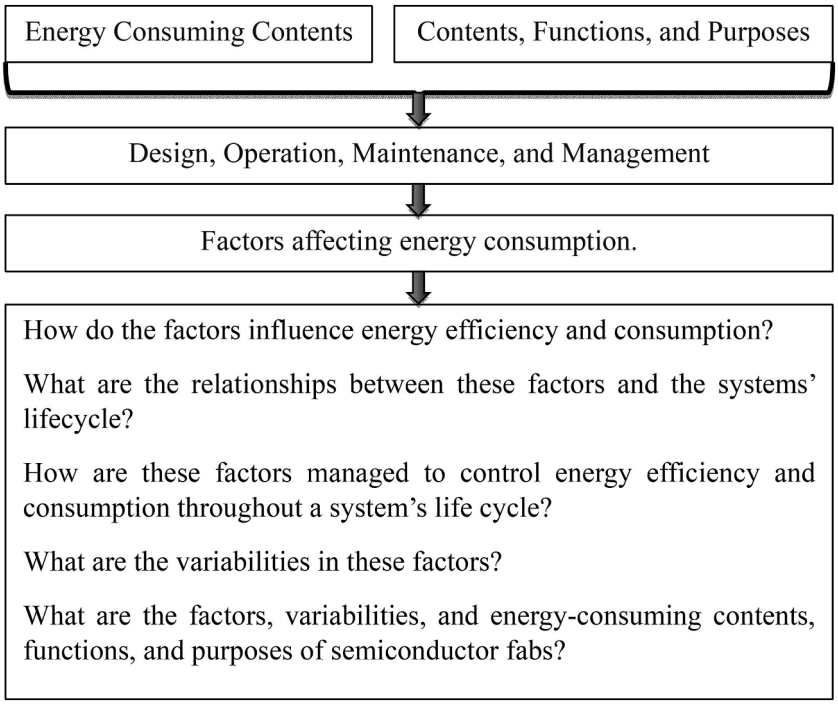


Fig. 1. Energy consumption, factors, and drivers of semiconductors.



Fig. 2. The image (left) shows a ducting system in a fab cleanroom and the HVAC system. The image (right) shows the fab cleanroom under installation.

trol, cooling or heating units at air vents, draft preventers, additional fans at vents, and air redistributors. More sophisticated equipment improves and optimizes HVAC systems. However, these functions satisfy different energy demands for various building types. Table 6 presents the differences between SMFs and RCIs.

Each equipment set has its own set of efficiency impediments. For example, more bends and junctions and installing higher-quality filters would reduce airflow speed, thus making the condenser and blower work harder (therefore increasing energy consumption). There are significant differences in how the RCI and SMF operate the equipment and the functions and purposes of the equipment. How would these differences affect the operation of SMF versus RCI?

HVAC functions depend on the occupants and the equipment housed in a building. HVAC systems' energy consumption in RCI heavily depends on the occupants and the appliances and equipment that the occupants use. In industrial manufacturing processes that require air quality that is higher standards than what occupants need, modeling occupancy would no longer be a practical approach to reduce energy consumption.

6. HVAC energy efficiency, management, and control strategies

HVAC systems for SMFs are designed to remove pointed heat load (at production source), microscopic size air particulates and toxic gases, and moisture injection into the indoor environment to prevent static electricity. SMFs HVAC systems focus on adjusting temperature and removing contaminants from outdoor air and making the air suitable for the indoor environment [73], overall equipment efficiency (e.g., bends and angles in ductwork, and equipment efficiency) [74], construction quality (e.g., system air leakages) [75], equipment position [76], outlets and inlets (design), and operation, maintenance, and operation procedures.

VAV and CAV systems also affect energy efficiency. CAV systems maintain equal air volumes, and a heating coil adjusts the outgoing air temperature [77]. VAV systems provide varying volumes of supply air according to space heating or cooling requirements [78]. According to a report by McKinsey, new SMFs are using VAV systems as they offer better performance, reliability, and sustainability.

Critical external factors directly affecting the energy consumed by HVAC systems are shown in Table 7, and the parameters are different from those for the internal factors in Table 8. Air was collected externally and treated before being distributed to the interior. The external air quality, temperature, and moisture content affect system operation. The purpose of different HVAC equipment is to maintain control of the temperature, relative humidity, radiant heat, thermal heat transfer, and air quality. The “Control Parameters” are what designers, engineers, property managers, and system manufacturers incorporate into their design, engineering, and management decisions. Tables 7 and 8, and 9A to 9C present the external, internal, and intrinsic (design) factors and their relationship with the control parameters with variably controlled units. Tables 9A–9C indicate the parameters and units for measurement of the factors.

Table 6
Individual components and how they operate in a typical HVAC system in SMFs.

Equipment/Functions	Differences between RCI and SMF
Furnace, boiler, and A/C (heat/cool)	RCIs address temperature variability caused by occupants, while SMFs extract heat from production.
Ventilator/Compressor/Fan	RCI's ventilator aims to meet fresh air supply for occupants while SMF's aims to bring sufficient cooled or heated air to the production.
Duct/Vent/Channel	RCI systems are primarily in aluminum or foil vents, while SMF systems can be both ducted and open ventilation systems without ductwork.
Thermostat	RCI's thermostat takes temperature throughout the buildings to meet occupants' needs, while SMF's thermostat targets production.
Filter	RCI's filter removes particulates that are larger than SMF's, and thus, more energy is needed to push air through the SMF's filters.
Humidifier/Dehumidifier/Condenser	RCI's occupants generate significant humidity, and balancing is the key challenge. SMF's production process is sensitive to electrostatics, so ensuring 50 % relative humidity is critical.
CAV control box	CAV meets the needs when the fresh air supply is more critical than temperature. Thus, CAV would better suit RCI.
VAV control box	VAV meets the needs of buildings where fresh air supply is less critical than temperature.

Table 7
Control parameters and variably controlled units for external factors.

Control Parameters on External Impact		
Factors	Control Parameters	Variables Controlled/Unit
Temperature	Fenestration, entrance, and setpoint temperature for heating and cooling.	°C (Temperature)
Humidity	Humidification or dehumidification systems.	% (Relative Humidity)
UV Ray	Filtration, ventilation, and building envelope.	Joule (Radiant heat)
Heat Island	Cooling systems, heat rejection, outdoor space design, building envelope, and process enclosures.	Joule (Radiant heat)
Building orientation and envelope	Insulation levels, window types, air leakage control, shading devices.	Watts (W) – Heat transfers through walls, roofs, windows, doors, and insulation
Air quality	Filter systems to extract particulates and toxins from the air.	Particles/ft^3 – Quantity and types of particles and toxins to remove

Table 8
Control parameters and variably controlled units for internal factors.

Internal Factors/Control Parameters		
Factors	Control Parameters	Variables Controlled/Units for Control
Occupants' Comfort	Psychrometric chart.	● °C (Temperature) ● Draft – Air flow rate: CFM or m^3/s (Ventilation)
Occupancy Classification	Occupancy counts, occupant types, and classification by NFPA 101.	● Total occupants and occupants' density. ● Proportion of transient and non-transient occupants. ● Occupant's types related to psychometrics.
Minor heat sources	Total heat emissions from minor sources.	● Joules (Heat) (minor heat sources – lighting, microwave, computers, etc.)
Building types	Cleanrooms, office spaces, laboratories, and production areas.	● ISO Class (Specific cleanroom classification requirements). ● People/ft^2 (Occupancy patterns and comfort requirements). ● °C, % RH, particles/m^3, lux, dB, or EMI, etc. (Specific environmental requirements for equipment performance and yield). ● Heat generating equipment and appliances. ● Air quality, humidity level, and airflow demand.
Functional performance	Energy consumption and efficiency metrics (Energy management systems that optimize the operation of HVAC systems).	● °C (Temperature control) ● m/s (Speed control) ● Particles/m^3 (Air quality) ● ISO Class (Cleanliness)
HVAC Machines and equipment	Chiller systems, AHUs [79], cooling towers, pump systems, exhaust systems, and process equipment.	● °C (Temperature control) ● % (Relative humidity control) ● m/s (Speed control) ● Pa (Pressure control) ● CFM or m^3/s (Airflow rates, gas flow rate) ● Unitless scale or % (Damper position control) ● Seconds (Process timing)
Equipment design	Temperature and humidity levels control size, capacity, heat transfer characteristics, and efficiency.	● tons or kW (Chillers) ● BTU/h or kW (Boilers) ● °C (Thermostats, AHUs – air-handling units) ● CFM or m^3/s (VAV systems, ventilation systems, and AHUs)

Fig. 3 highlights the key external and building factors that SMFs should focus on and the key performance indicators that SMFs have to track. Energy efficiency should be achieved in SMFs without sacrificing building function and purposes – ***the production of chips***. External and building factors (EFs and BFs) are the critical factors incorporated by energy models, codes, standards, and best practices. The importance and relevance of these factors towards the functions and purposes of a building must be identified. Designers, engineers, and property managers should focus on fundamental engineering factors (ten items listed) to design HVAC systems in SMFs and align them with key performance indicators to achieve optimization and energy efficiency at the same time [80–82] despite prior studies highlighting that systems efficiency is affected by installation [83]. Clients often influence decisions and choices, impacting system performance and efficiency [84].

Moreover, the targeted performance indicators from system design based on engineering factors would generate more energy consumption data, which could be collected to inform future design considerations and thus influence engineering decisions as a closed-loop process. Therefore, it is crucial to explore the relationships between design, engineering, and consumption data for system optimization by appropriate modeling methods that always need intense data from the abovementioned ones. That will be further addressed later.

To fulfill the optimization of systems in SMFs, other than collecting intense data, the modeling methods to be used might be different due to various considerations such as systems, design-system interface, building elements, external factors, and interior. The systems encompass multiple technologies, from cleanroom design to manufacturing equipment. The design-system interface is critical for efficient integration and workflow. Building elements are designed following the stringent requirements of semiconductor manufacturing. External factors also influence the design to ensure stability and reliability. Interior factors include the layout, organization, and cleanliness standards necessary for maintaining an uncontaminated and controlled manufacturing environment.

These meticulous considerations in the modeling process allow for a comprehensive understanding of energy dynamics in facilities. Sophisticated modeling methods enable precise predictions of energy usage under varying conditions. Since modeling is a dynamic tool for SMFs' cleanrooms to navigate energy codes and standards, HVAC systems are compliant and optimized for efficient and sustainable operation.

7. Energy codes and standards for HVAC systems in semiconductor cleanrooms

Many codes, standards, guidelines, and practices are relevant to cleanroom design, construction, maintenance, and management, even though many are developed for residential and commercial buildings. Semiconductor fabrication plants have unique energy requirements that need to be adequately addressed by existing energy codes and standards due to the complexity and variability of the

Table 9A
Control parameters and variably controlled units for intrinsic factors.

Design and Intrinsic Factors/Control Parameters		
Factors	Control Parameters	Variables Controlled/Unit
System design	Integrated filtration, optimized air distribution and ventilation, process exhaust and make-up air requirements, layout, and zoning.	Load calculation: ● BTU/h or kW (Heat load) ● CFM or m^3/s (Airflow) Ductwork: ● inches or mm (Duct size and piping diameter) ● CFM or m^3/s (flow rates) Others: ● ft^2 or m^2 (Zone area) ● °C (Temperature control) ● % (Energy efficiency) ● $ (Cost constraints)
Building design	Appropriate insulation and sealing, controlled air infiltration, cleanroom classification, and other semiconductor facility requirements.	HVAC and mechanical systems integration: ● °C (Temperature) ● CFM or m^3/s (Airflow) Electrical and lighting design: ● Watts or kW (Power) ● Lux (Illuminance) Interior design: ● ft^2 or m^2 (Area) Safety: ● hours (Fire rating) Renewable energy: ● kWh or MW (energy generation) Water efficiency: ● GPM or L/s (Water flow) ● gallons or liters (water consumption)
Fenestration	Window types and shading devices.	Size and placement: ● ft, inches, m, cm (dimensions) Building codes: ● Code sections, paragraphs, or articles Safety: ● hours (Fire rating) Air leakage: ● CFM or m^3/s (Air leakage rate) Interior design: ● ft^2 or m^2 (Area) ● ft, inches, m, cm (Furnishing dimensions)

industry [85]. Future research should focus on integrating and modifying these codes, standards, guidelines, and practices relevant to cleanroom design, construction, maintenance, and operation. These codes and standards include.

7.1. Energy design standards

ASHRAE is the American Society of Heating, Refrigerating, and Air-Conditioning Engineers. ASHRAE standard 90.1 aims at energy optimization [86]. The codes benchmark energy performance and guide the best practices for building systems design and operation [87]. One of the significant changes is the Mechanical System Performance Path with Total System Performance Ratio (TSPR) metric, an energy-efficient approach within the building design and HVAC [88]. TSPR provides flexibility during design while optimizing HVAC systems. Thus, TSPR would benefit the energy design path for SMFs. The International Energy Conservation Code (IECC) [89] is similar to ASHRAE 90.1 (minimum energy efficiency standards) [90].

7.2. Cleanroom standards

ASHRAE Standard 170 is a cleanroom design and construction guideline where the focus is removing pollutants and particulates for health, safety, and manufacturing [91] where healthcare facilities, pharmaceutical, electronics, and semiconductor industries rely on to produce high-quality and contamination-free environments (from patients to manufacturing). These standards include filter, airflow, and equipment energy consumption standards.

7.3. Indoor air quality standards

ASHRAE 62.1/62.2 addresses indoor air quality in buildings [92]. The standard focuses on removing poor-quality air and increasing fresh air supply; however, the focus is on occupants and not the manufacturing process. The ventilation principles, nevertheless, would be applicable in cleanrooms.

Table 9B
Control parameters and variably controlled units for intrinsic factors.

Design and Intrinsic Factors/Control Parameters		
Factors	Control Parameters	Variables Controlled/Unit
Entrance	Airlock systems.	Location and placement: ● degrees, minutes, seconds (Latitude and Longitude) Security: ● ft^2 or m^2 (Secured area) Accessibility: ● inches or cm (Clear width) Entrance space: ● ft, inches, m, cm (Clearance height) Intercom system: ● ft, m (communication range)
Operation management	Temperature, humidity, and air quality parameters, integrated energy management systems for real-time monitoring and optimization.	Scheduled maintenance: ● days, weeks, months, years (Time intervals) Energy efficiency measures: ● kWh, % (Energy savings) Safety: ● ft, m (Emergency exits) Indoor air quality: ● ACH (air exchange rate) Financial management: ● $ (Currency) Building automation systems: ● °C, % (Control setpoints) Code compliance: ● Code sections, paragraphs, or articles
Maintenance	Inspection and calibration, replacement of inefficient HVAC components, and the cleaning of filters.	Scheduled maintenance: ● days, weeks, months, years (Time intervals) Inventory management: ● units, pieces (Quantities) Labor and outsourcing: ● hours (Labor hours) Documentation: ● pages (Page count) Cost control: ● $ (Currency) Regulations: ● Code sections, paragraphs, or articles Skill development: ● hours (Training hours)

7.4. Energy efficiency guidelines

Energy Star is a series of energy efficiency guidelines developed by the Environmental Protection Agency (EPA) and the Department of Energy (DOE). The guidelines provide information on the energy consumption of products and devices using various standardized methods, and the program helps consumers identify energy-efficient products and devices that save energy and money [93]. Energy Star HVAC systems would save more energy than non-Energy Star.

7.5. Energy efficiency program

The U.S. DOE's Building Technologies Office (BTO) program focuses on advancing high performance, energy efficiency, and decarbonizing/carbon-neutral technologies, techniques, tools, and services in residential and commercial buildings. The program's key areas include the development of innovative, cost-effective energy solutions, R&D, market stimulation, and building codes and equipment standards [94,95].

Tables 10A–10C organize these codes and standards relevant to SMFs' HVAC systems.

8. Data and energy efficiency for SMFs' HVAC systems – integrating production with energy design, installation, and operation

The energy consumption pattern for HVAC systems in SMFs is less dynamic than RCI's due to the lack of changing occupancy. 24/7 manufacturing operation and the changing outdoor environment shielded by a thick façade increase the stability of energy consumption. While most of the SMFs' energy-consuming processes are less dynamic, changing outdoor air conditions and production variation would affect the dynamism of energy consumption. Using the three BFs, two EFs, and ten ENGFs from Fig. 3, this section develops

Table 9C
Control parameters and variably controlled units for intrinsic factors.

Design and Intrinsic Factors/Control Parameters		
Factors	Control Parameters	Variables Controlled/Unit
Replacement	Retrofitting systems, upgraded equipment, and advanced technology to improve efficiency.	● number of inspections (Inspections) ● Scales or metrics (Risk assessment) ● Project's life cycle (Assess lifecycle management) ● $ (Budget) ● kWh, BTUs, % (Energy efficiency, energy consumption reduction percentages) ● days, weeks (Scheduling, installation, and commissioning)
Design	AutoCAD, BIM models.	● Scale ratios (1:25, 1:50, etc. for scale and proportion) ● ft, inches, m, cm (Layout dimensions, clash detection) ● ft^2, ft^3, m^2 (Geometry shapes) ● pixels (Visualization resolution) ● pages (documentation page count)
Equipment specification	High energy efficiency ratings for selected equipment, compliance with industry standards and regulations specific to semiconductor facilities.	● kWh, % (Energy efficiency/savings) ● tons or kW (Capacity like cooling) ● ft, inches, m, cm (Size dimensions) ● Code sections, paragraphs, or articles (Regulatory compliance) ● days, weeks, months, years (Service and maintenance time interval) ● $ (Budget currency) ● % on/off, etc. (Automation control setpoints or levels) ● bps, Mbps, etc. (Data transfer rate for data communication)

a data-intensive energy design framework for SMFs that focuses on optimizing production by integrating HVAC energy efficiency technical and engineering performances and indicators and energy management and control strategy into the framework. The three engineering criteria driving HVAC system's energy consumption in SMFs include.

8.1. The first criteria

The chip production process requires specific air conditions for an optimum chip production environment.

- Air quality requires particles smaller than 5 μm for ISO 9 cleanroom and between 1 and 2 μm for ISO 1 rated cleanroom to be removed.
- Heat removal (at optimum and idling): Heat generated in SMFs is mainly concentrated along specific points in the production line. The workers stationing in the fabs must wear protective clothing, and the ambient temperature would usually be 19–23 °C in the fab.
- Moisture injection (at optimum and idling): SMFs must always maintain a 35–65 % relative humidity to avoid static electricity.
- Production disruption and reducing production below the optimal levels would decrease energy use but would sacrifice energy use efficiency.

8.2. The second criteria

The outdoor environment constantly changes, and the condenser, compressor, make-up air unit, and air handling unit adjust their operations and efficiency to meet indoor air standards. Air particles smaller than 0.1 μm are removed, cooled, or heated to 19–23 °C, and moisture is removed or injected to 50 % relative humidity before the air is distributed into the fab. Indoor air has to be returned to a similar quality and redistributed to the factory. Recycled air from the fab is contaminated with chemicals used in chip manufacturing. All particles and contaminants must be removed before air is heated or cooled and moisture is injected or removed.

8.3. The third criteria

The HVAC system technical and engineering performances and energy design models are based on codes, standards, best practices, specifications, and criteria including (a) sizing of pipe, ductwork, and equipment sizing; (b) shapes, bends and curves of air and liquid deliver and distribution systems; (c) power distribution and generation of systems; (d) equipment and outlet positioning; (e) equipment and System types; (f) indoor Environmental performance; (g) particulates allowed in atmosphere (indoor and disposal); (h) resistance and friction within equipment and along systems; (i) inefficiencies of individual equipment and throughout the system due to underutilization. Each component of the system performs specific functions.

a. Removal of particles < 0.1 μm.
b. Removal of chemical contaminants in the air.

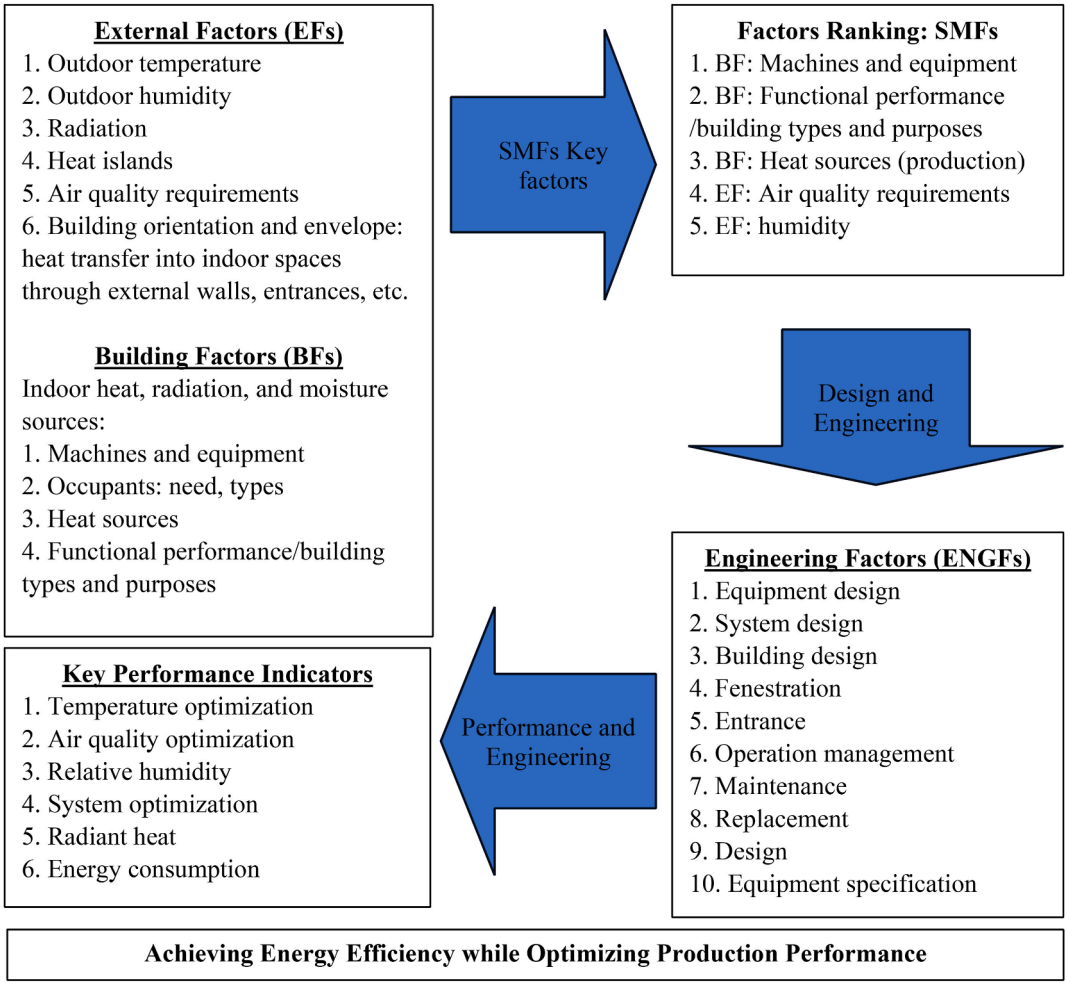


Fig. 3. Relationships between external, building, engineering factors, and performance indicators.

Table 10A
The codes, standards, and designs are incorporated into SMFs.

Organizations/Institutions	Codes/Standards/Documents	Descriptions
The Associated Air Balance Council (AABC)	National Standards for Total System Balance	Comprehensive manual details of the minimum standards for total system balance.
Air Conditioning and Refrigeration Institute (AHRI)	885	To apply air terminals and air outlets, estimate sound pressure levels in a conditioned space.
	575	Sound pressure measurement.
American National Standards Institute (ANSI)/AHRI-550/590	550/590	Chiller system testing.
ANSI/Underwriters Laboratories (UL)	465	Central cooling air conditioners.
Air Movement and Control Association (AMCA)	210	Laboratory method of testing fans for rating.
	300	Reverberant room method for sound testing of fans.
	301	Method of publishing sound ratings for air-moving devices.

c. Cool or heat air to 19–23 °C.
d. Remove or inject moisture to achieve 50 % relative humidity.
e. Overcoming friction in ducts, filters, bends, surfaces, registers, outlets, inlets, etc.
f. Overcoming system operational efficiency due to operation at above or below optimal operation.
g. Overcoming and eliminating air draft.
h. Equipment conditions.
i. Equipment and system sizes and optimal operation conditions.

Table 10B
The codes, standards, and designs are incorporated into SMFs.

Organizations/Institutions	Codes/Standards /Documents	Descriptions
American Society of Heating, Refrigerating, and Air-Conditioning Engineers (ASHRAE)	HVAC Application Handbook	Noise and vibration control.
	Standard 62.1	Ventilation for acceptable indoor air quality.
	(ANSI included) 90.1	Meet energy conservation in new building design (fan/motor efficiency, etc.).
International Energy Conservation Code (IECC)	–	Similar to ASHRAE 90.1.
American Society for Testing and Materials (ASTM)	A90	Test method for the weight of coating on Zinc-coated (galvanized) Iron or Steel articles.
	A167	Specification for stainless and heat-resisting chromium-nickel steel plate, sheet, and strip.
	A527	Specification for general requirements for steel sheet, zinc-coated (galvanized) by the hot-dip process, and lock-forming quality.
	A653	Specification for steel sheet, Zinc-coated (galvanized) by the hot-dip process.
	A924	Specification for general requirements for sheet steel, metallic-coated by the hot-dip method.
	D6386	Practice for preparation of Zinc (hot dip galvanized) coated Iron and Steel products and hardware surfaces for painting.
	D2513	A standard specification for thermoplastic gas pressure pipe, tubing, and fittings.
American Society of Mechanical Engineers (ASME)	B31.3	Process piping.
	BPVC	Boiler and pressure vessel code.
	B36.10	Wrought Steel pipe schedule (SCH).
	B36.19	Stainless Steel pipe nominal wall thickness designation.
American Welding Society (AWS)	A2.4	Standard symbols for welding.
	412	Installing seismic restraints for mechanical equipment.
Federal Emergency Management Agency (FEMA)	414	Installing seismic restraints for duct and pipe.
International Mechanical Code (IMC) 2018 and Phoenix Amendment 2018	–	Provide prescriptive requirements and performance-related requirements for mechanical systems.
National Environmental Balancing Bureau (NEBB)	–	Procedural Standards for testing, adjusting, and balancing (TAB) environmental systems.

Table 10C
The codes, standards, and designs are incorporated into SMFs.

Organizations/Institutions	Codes/Standards/Documents	Descriptions
Sheet Metal and Air Conditioning Contractors' National Association (SMACNA)	Duct design	Describe how HVAC air distribution systems are designed using basic methods and procedures.
	Commissioning manual	Describe the knowledge of commissioning for new buildings and re-commissioning for existing buildings.
	Duct leakage test manual	Describe the proven methods for testing ductwork for air leaks.
	Duct construction standards	Represent the state of the art in sheet metal duct fabrication and installation requirements.
	Systems inspection	Provide the guidelines for the inspection of commercial HVAC duct systems.
	Systems testing, adjusting, and balancing	Describe the testing, adjusting, and balancing (TAB) of all HVAC systems in a new building needed to complete the installation and to make the systems perform as the designer intended.
	Seismic restraint manual guidelines for mechanical systems	Identify the correct restraints for sheet metal ducts, pipes, and conduits so that they are more likely to remain attached to the building during an earthquake.

Friction (caused by ductwork, bends, connectors, registers, outlets, inlets, etc.), operational procedures, air draft, equipment conditions, poor sizing, and poor maintenance, construction, and installation would affect HVAC system performance and efficiency, failing to meet specifications as well.

Fig. 4 summarizes the above discussions.

9. Relationships between HVAC designs, engineering, and data – the path to use artificial intelligence HVAC systems design, construction, installation, maintenance, and operation (DCIMO)

As discussed earlier, the research addressed the imperative of integrating data throughout the entire lifecycle of HVAC systems in SMFs — from the design and installation phases to daily operations. The relationships between HVAC designs, engineering principles, and the vast data available at the design, installation, and maintenance/operation phases enable a transformative evolution in the field – using Artificial Intelligence (AI) to narrow the gap between design, installation, and operation. Fig. 5 illustrates a framework

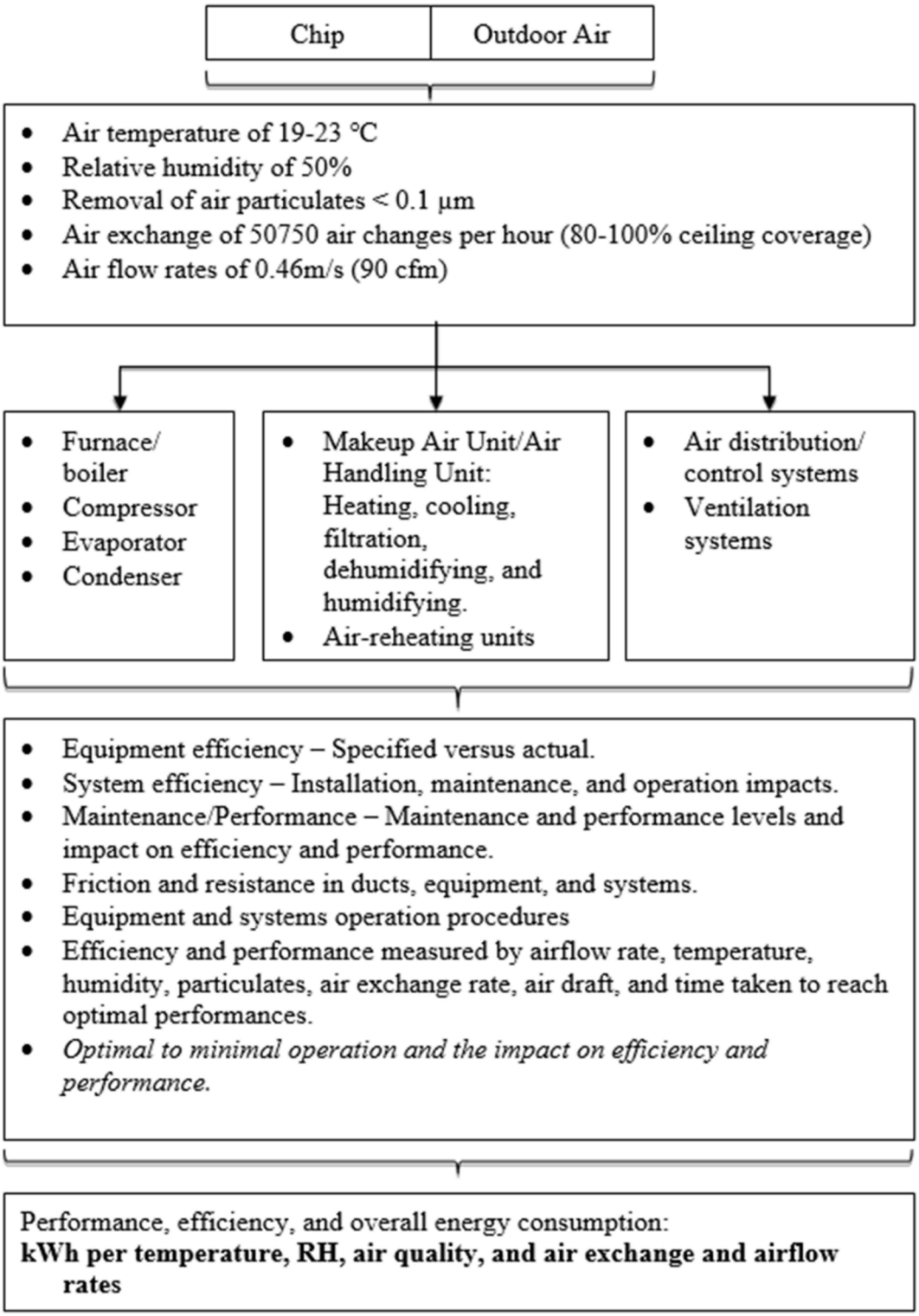


Fig. 4. Functional performances, equipment and systems, and control parameters of HVAC systems in SMFs.

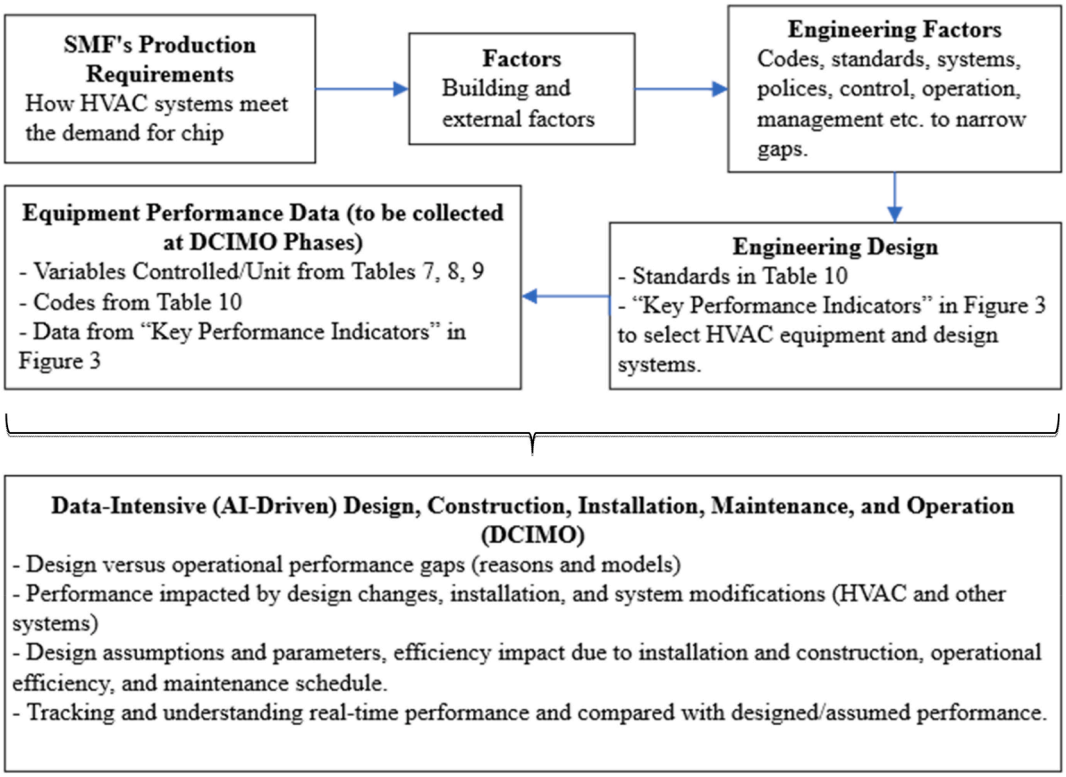


Fig. 5. Framework to intensify the use of data for DCIMO of HVAC systems.

for utilizing AI. The key to using AI in HVAC design, installation, and maintenance/operation circles around the use of data. First, the requirements, performance, and indicators must be identified. These include.

- The relationships between the SMF production requirements and the required performance of HVAC systems would determine the required system performance to achieve optimal production (discussed in previous sections).
- Codes, standards, systems, policies, control, operation, and management requirements for HVAC systems would translate the requirements into actual performance indicators for HVAC systems.
- The key performance indicators for the system's overall performance (air quality, quantity, temperature, and humidity).
- Individual unit and equipment performance indicators (e.g., CFMs for air intake, equipment or system efficiency, bends and curves in ducts that reduce efficiency, etc.).

Data would be collected from the following.

- The equipment and system performances (tracking using their performance indicators) to measure the equipment and system efficiencies and ability to comply with the required equipment and system performances.
  a. System performances collected at selected multiple points:
     - Air temperatures.
     - Air quality monitoring data using air quality monitoring systems.
     - Airflow/draft.
     - Relative humidity.
     - Specifications versus tests after installation, at commissioning, and during operation.
  b. Equipment and components' performance:
     - Individual equipment specifications versus actual output performance.
     - Airflow and temperature losses before and after bends, angles, and curves (changes during design and installation).
     - Inefficiencies between equipment.
     - Specifications compared to, first, at installation, second, at commissioning, third, during operation, and fourth, after maintenance.
  c. BIM objects and models for the equipment and system to track changes in the design and changes to the specified and actual performances using the performance indicators.

The ultimate goal of the exercise is to establish the relationships between the performance indicators and design and use machine learning to understand how design changes (BIM objects and models) would affect the actual equipment and system performances.

10. Conclusions and future research

This study identifies critical energy consumption factors specific to semiconductor manufacturing facilities (SMFs) and analyzes how these factors influence system performance and impact HVAC system optimization. Our developed framework integrates design engineering and energy consumption data, providing a foundation for an AI-based, data-intensive energy design approach tailored to SMFs. This approach allows for the effective collection and analysis of performance data, significantly influencing energy design.

The recommendations aim to replace outdated and generalized energy codes and standards with a novel HVAC energy-design framework. This framework is precisely engineered to meet the unique energy demands of SMFs, promoting efficiency and sustainability in semiconductor manufacturing. It represents a shift towards a more customized and precise approach to HVAC system optimization in industrial settings.

The study proposes further research to refine and expand this framework. The next phase will focus on establishing models that connect various stages of design, construction, installation, maintenance, and operation of HVAC systems by integrating Generative AI. This will bridge the current gaps and enhance performance while maintaining energy efficiency at an optimal level.

By leveraging AI and comprehensive data analytics, future research will continue innovating and improving HVAC systems' operational standards and sustainability in high-tech industries, particularly semiconductor manufacturing. This progress is essential for reducing energy waste and minimizing greenhouse gas emissions, thereby contributing to global environmental sustainability goals.

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CRediT authorship contribution statement

Hsiao-Ping Ni: Writing – original draft. **Wai Oswald Chong:** Writing – review & editing. **Jui-Sheng Chou:** Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Hsiao-Ping Ni reports financial support was provided by United Integrated Services. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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